Functional load modulates acoustic and articulatory variability in English

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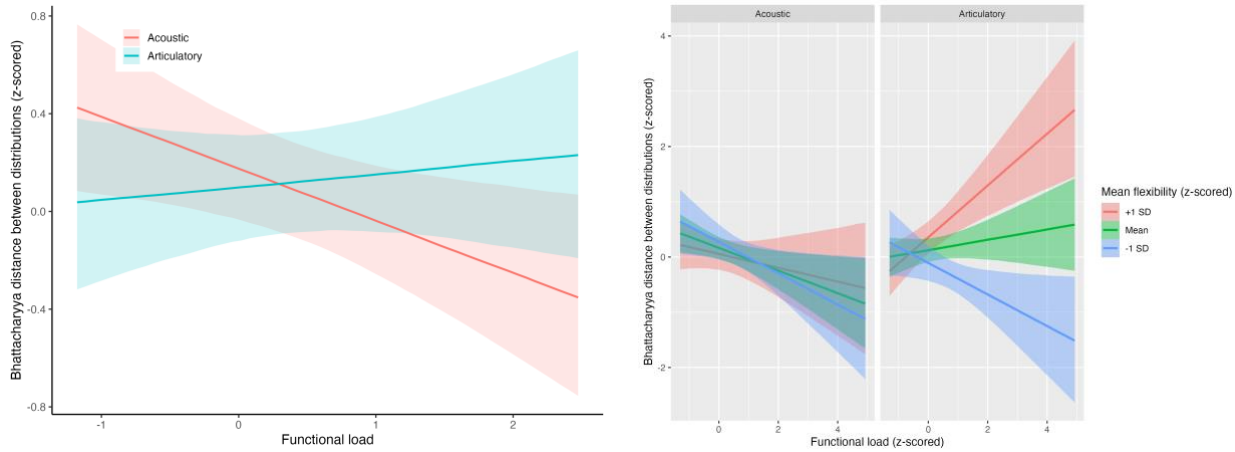
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Introduction. Variation in speech production has been shown to be modulated by various factors. This study focuses on information-theoretic considerations (Shannon, 1948), which have been found to predict variability in speech (Baker & Bradlow, 2009, *inter alia*). Despite the general finding that perceptual recoverability pressures lead to lower variability on items that are more informative, consideration of articulatory efficiency, which predicts higher variability due to coarticulation, has also been put forth (Xu & Prom-On, 2019). In a parallel piece of literature, the multiple-to-one relation between articulatory synergies and the speech target suggests that certain flexibility could exist between articulation and acoustics (Perrier & Fuchs, 2015), further relaxing the tension of variability between the two domains. This study combines both aspects and examines variability between phoneme pairs in English. Through unsupervised examination of an ultrasound corpus, we show such functional effects. Such conditioning, however, was in the opposite direction to the prediction.

Methods. Following Huang & Breiss (2026), we measured the amount of variability for a phoneme pair as the distance between the two phonemes' respective distributions in the acoustic and articulatory spaces. We used ultrasound/optical lip imaging data and acoustic data from the TaL Corpus (Ribeiro *et al.*, 2020). The imaging and acoustic data were mapped to the corresponding articulatory/acoustic spaces through 3D-/1D-CNN autoencoders. The acoustic data was first converted to MFCCs. The autoencoders were then trained to map each video clip/MFCC vector to a 1024D latent space. The decoder component then learned to reconstruct the original data based on the latent vector. After training, the latent spaces then served as the articulatory/acoustic spaces. The flexibility of each phoneme was taken as the amount of uncertainty of the articulation, given the acoustics. This was estimated as the conditional entropy of the articulatory vector given the acoustic vector through a mixture-density network. The functional load of a phoneme pair was calculated as the perceptual confusability-based entropy difference (Burchill & Jaeger, 2017) in the British National Corpus (BNC Consortium, 2007). In this study, we examined all possible phoneme pair combinations among consonants and vowels. Finally, the amount of variability for a phoneme pair was taken as the Bhattacharyya distance between the distributions of all tokens of the two phonemes.

Results. Overall, distinctiveness was modulated by the functional load of the pair. However, while functional load increased distinctiveness in articulation, the reverse was found in acoustics. Flexibility, on the flip side, increased the positivity of the functional load effects across both modalities.

Discussion. Our results support previous studies where information-theoretic correlates predict speech variation. The positive effects of functional load in articulation indicate orientation toward perceptual recoverability, though such effects were reversed in acoustics. This result suggests that perceptual recoverability plays an important role in articulation as well. The null functional load effects in acoustics, on the flip side, may be reconciled by positing that the variability in acoustics is already sufficiently minimized, and functional pressures consequently have weaker effects. This ceiling effect could be further supported by the interaction between functional load and flexibility: In both domains, flexibility further enhanced the functional load effects in the positive direction, but such enhancement was limited in acoustics. Such a hypothesis requires further investigation.



(a) (b)

Figure 1: *The interaction between the distance between the two phonemes' respective distributions and modality, functional load, and phoneme flexibility.*

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